Wave Function of the Void: Dual-Sector Quantum Cosmology on S^3, Emergent Time, Entropy, and Geometric Expansion

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Abstract. We present a mathematically rigorous dual-sector quantum cosmology in which the Hilbert space of the Universe factorises as $\mathscr{H} = \mathscr{H}_{\rm m} \otimes \mathscr{H}_{\rm v}$, with $\mathscr{H}_{\rm v}$ carrying a *Wave Function of the Void* $\Psi_{\rm v}$. Starting from the full ADM action on a compact, orientable three-manifold, we derive the void-extended Wheeler–DeWitt (WDW-V) constraint. The void sector is modelled as a *standard* (non-phantom) harmonic-oscillator field with positive kinetic term and positive-definite zero-point energy. Entanglement between the matter and void sectors, generated by a minimal bilinear coupling $\hat{H}_{\rm int} \propto a^3 \hat{\phi}_m^2 \hat{\phi}_v^2$, drives a monotonically increasing von Neumann entropy $S_{\rm ent}(a)$, providing a first-principles derivation of the second law of thermodynamics from wave-function entanglement alone, with no low-entropy boundary condition imposed by hand. A WKB analysis of the gravitational sector yields an emergent time parameter t via the Page–Wootters mechanism, and the arrow of time follows from the asymmetric spectra of the two sectors. We prove that normalisability of $\Psi_{\rm v}$ together with simple connectivity uniquely selects the three-sphere S^3 as the spatial topology via the Poincaré–Perelman theorem, eliminating the initial-singularity problem. *Crucially, the observed accelerated recession of galaxies is re-interpreted geometrically:* since the spatial volume of S^3 scales as $V(a) = 2\pi^2 a^3$, a linear change in the curvature radius produces a cubic change in volume and a proportional stretching of geodesic distances $d = a\chi$. No dark-energy field, phantom sector, or negative pressure is required. Observational signatures include characteristic oscillations in the CMB power spectrum and a purely geometric redshift–distance relation.

Keywords: quantum cosmology; Wheeler–DeWitt equation; entanglement entropy; arrow of time; Page–Wootters mechanism; three-sphere topology; geometric expansion.

Contents

1 Introduction and Motivation **4**

1.1 The three unsolved puzzles . . . 4
1.2 The void as an active partner . . . 4
1.3 Geometric origin of the apparent expansion . . . 4
1.4 Relation to prior work . . . 5

2 Hilbert Space Structure **5**
2.1 The bipartite decomposition . . . 5
2.2 The total state and Schmidt decomposition . . . 6
2.3 Reduced density matrices . . . 6

3 The Hamiltonian and the WDW-V Equation **6**
3.1 ADM action on S^3 . . . 6
3.2 Canonical quantisation — void as a standard scalar sector . . . 6
3.3 The Wheeler–DeWitt–Void equation . . . 7
3.4 The void Hamiltonian . . . 7
3.5 The interaction Hamiltonian as generator of entanglement . . . 7

4 Entanglement Entropy and the Second Law **8**
4.1 Von Neumann entropy of the matter sector . . . 8
4.2 Monotonic growth: proof of the second law . . . 8
4.3 Entropy as a function of the scale factor . . . 9

5 Emergent Time and the Arrow of Time **9**
5.1 WKB factorisation and emergent time . . . 9
5.2 Schrödinger equation with emergent time . . . 9
5.3 The phase differential and the arrow of time . . . 10
5.4 The Page–Wootters mechanism . . . 10

6 Topological Structure: The Three-Sphere **11**
6.1 Necessity of compact, simply connected spatial sections . . . 11
6.2 The metric of S^3 and the cosmic scale . . . 11
6.3 Absence of initial singularity . . . 12
6.4 Poincaré recurrence on S^3 . . . 12

7 Geometric Expansion: From S^3 Curvature to Observed Recession **12**
7.1 Observer as a geodesic neighbourhood . . . 12
7.2 Cubic volume scaling . . . 13
7.3 What an internal observer sees . . . 13
7.4 Energy density and the scaling law $\rho \propto 1/a^3$ . . . 13
7.5 Why the observer perceives acceleration . . . 14
7.6 Limiting regimes of S^3 geometry . . . 15

7.7 No need for dark energy 16

8 Emergent Friedmann Dynamics on S^3 16

8.1 Born rule for geometry 16

8.2 Effective Friedmann equation 17

8.3 Hubble rate and geometric deceleration/acceleration 17

9 Observational Signatures 18

9.1 CMB power-spectrum oscillations 18

9.2 Geometric redshift–distance relation 18

9.3 Quantum-gravitational correction to the Hubble rate 18

10 Discussion 19

10.1 Comparison with existing frameworks 19

10.2 Information and unitarity 19

10.3 Distinguished value in the Schmidt spectrum 19

11 Conclusions 19

A DeWitt Measure and Factor Ordering 20

B Schmidt Decomposition and Two-Mode Approximation 21

C WKB Solution and Semiclassical Regime 21

D Entropy Monotonicity: Full Proof 21

E Golden-Ratio Structure in the Schmidt Spectrum 21

1 Introduction and Motivation

1.1 The three unsolved puzzles

Standard quantum mechanics provides an extraordinarily successful description of microscopic phenomena, yet its application to the Universe as a whole uncovers three deep and interrelated paradoxes.

(i) The reversibility paradox. The Schrödinger equation $i\hbar\, d_t|\Psi\rangle = \hat{H}|\Psi\rangle$ is time-reversal invariant, yet the macroscopic world exhibits a definite arrow of time. Boltzmann's H-theorem requires a low-entropy initial state as an *external* postulate.

(ii) The initial-condition enigma. Penrose's Weyl-curvature hypothesis [1] quantifies the extraordinary improbability of the Big Bang initial state, $\Omega_{\rm init}/\Omega_{\rm max} \approx e^{-10^{123}}$. No dynamical mechanism within standard cosmology explains this fine-tuning.

(iii) The singularity dilemma. Einstein's field equations predict a curvature singularity at $t = 0$. The standard Wheeler–DeWitt equation [2, 3] preserves the singularity at $a = 0$ unless additional structure is imposed.

1.2 The void as an active partner

We contend that all three paradoxes share a common origin: the wave function of the Universe has been truncated. Standard quantum cosmology assigns a wave function only to the matter–geometry sector. The vacuum — the *void* — is treated as a passive backdrop. We propose that the void carries its own dynamical wave function $\Psi_{\rm v}$, encoding the latent potentiality of empty space. The void is modelled as an ordinary quantum sector with a positive kinetic term; no exotic "phantom" ingredient is invoked. The dual structure restores unitarity while simultaneously generating:

(a) a monotonically growing entanglement entropy explaining the second law;

(b) an emergent time and arrow fixed by the relative phase of the two sectors;

(c) a topological constraint that selects S^3 and removes the singularity.

1.3 Geometric origin of the apparent expansion

A distinguishing feature of this paper is that we do not invoke a dark-energy field, a cosmological constant sourced by vacuum energy, or any negative-pressure component to explain the observed recession of galaxies and the late-time acceleration. Instead, we take seriously the compact three-sphere topology of space: since

$$V(a) = 2\pi^2 a^3, \qquad d = a\chi, \qquad \lambda \propto a,$$

any change in the curvature radius a stretches geodesic distances and light wavelengths in direct proportion, and the cubic scaling of the volume automatically makes the apparent growth of separations super-linear in the change of a. The observed expansion is real from the observer's point of view but geometric in origin — a direct consequence of living inside a geodesic neighbourhood $B^3 \subset S^3$ whose global curvature radius evolves.

1.4 Relation to prior work

Our framework builds on, but is distinct from, several existing proposals. The Hartle–Hawking no-boundary proposal [4] introduces a specific wave function for the gravitational sector; we augment this with an independent void sector. The Page–Wootters mechanism [5, 6] derives time from entanglement between a "clock" and a "system"; we apply this insight to the matter–void bipartition. Unlike phantom dark-energy models [7], we do not employ negative kinetic terms: the void sector has standard, positive-definite kinetic energy. Conformal cyclic cosmology [8] proposes cosmic recurrence; our S^3 topology provides a topological basis for recurrence via the Poincaré–Perelman theorem [9–11].

Conventions. We use natural units $c = 1$. The reduced Planck mass is $m_{\mathrm{Pl}} = (8\pi G)^{-1/2}$. Signature $(-,+,+,+)$. Bold/hatted symbols denote operators.

2 Hilbert Space Structure

2.1 The bipartite decomposition

Let the total Hilbert space of the Universe be

$$\mathscr{H} = \mathscr{H}_{\mathrm{grav}} \otimes \mathscr{H}_{\mathrm{m}} \otimes \mathscr{H}_{\mathrm{v}}, \tag{1}$$

where $\mathscr{H}_{\mathrm{grav}} = L^2(\mathbb{R}_{>0}, \mathrm{d}\mu_{\mathrm{DeW}})$ is the minisuperspace Hilbert space with DeWitt measure $\mathrm{d}\mu_{\mathrm{DeW}} = a\,\mathrm{d}a$ (Appendix A), $\mathscr{H}_{\mathrm{m}}$ is the Fock space of matter fields, and $\mathscr{H}_{\mathrm{v}} = L^2(\mathbb{R}, \mathrm{d}\phi_v)$ is the void sector.

For entropy and time emergence we use the bipartite decomposition

$$\mathscr{H} = \mathscr{H}_{\mathrm{grav+m}} \otimes \mathscr{H}_{\mathrm{v}} \equiv \mathscr{H}_A \otimes \mathscr{H}_B. \tag{2}$$

Definition 2.1 (Wave Function of the Void)**.** The *Wave Function of the Void* is the element $\Psi_{\mathrm{v}} \in \mathscr{H}_{\mathrm{v}}$ defined by $\Psi_{\mathrm{v}}(a, \phi_v) := \langle a, \phi_v | \Psi_B \rangle$, where $|\Psi_B\rangle$ is the reduced state of the void sector.

2.2 The total state and Schmidt decomposition

In the two-mode Schmidt truncation (justified in Appendix B),

$$|\Psi(a)\rangle = \cos\theta(a)\ |\psi_m(a)\rangle \otimes |0_v\rangle + \sin\theta(a)\ |0_m\rangle \otimes |\psi_v(a)\rangle\,, \tag{3}$$

with $\theta(a) \in [0, \pi/2]$ the Schmidt angle. The normalisation $\cos^2\theta + \sin^2\theta = 1$ is automatic.

2.3 Reduced density matrices

$$\hat{\rho}_m(a) = \mathrm{tr}_v\,|\Psi\rangle\langle\Psi| = \cos^2\theta\ |\psi_m\rangle\langle\psi_m| + \sin^2\theta\ |0_m\rangle\langle 0_m|\,, \tag{4}$$

$$\hat{\rho}_v(a) = \mathrm{tr}_m\,|\Psi\rangle\langle\Psi| = \cos^2\theta\ |0_v\rangle\langle 0_v| + \sin^2\theta\ |\psi_v\rangle\langle\psi_v|\,. \tag{5}$$

Both are valid density matrices.

3 The Hamiltonian and the WDW-V Equation

3.1 ADM action on S^3

Starting from the ADM action on a compact spatial three-manifold,

$$S = \frac{m_{\rm Pl}^2}{2}\int \mathrm{d}^4x\sqrt{-g}\,R + S_m[\phi_m, g_{\mu\nu}] + S_v[\phi_v, g_{\mu\nu}] + S_{\rm GH}, \tag{6}$$

with the FRW ansatz

$$\mathrm{d}s^2 = -N^2(t)\,\mathrm{d}t^2 + a^2(t)\,\mathrm{d}\Omega_3^2, \qquad \mathrm{d}\Omega_3^2 = \mathrm{d}\chi^2 + \sin^2\chi\,\mathrm{d}\Omega_2^2, \tag{7}$$

and $V_3 = 2\pi^2$ the dimensionless volume of the unit S^3. The reduced action becomes

$$S = \int \mathrm{d}t\ N\left[-\frac{3\pi m_{\rm Pl}^2}{N^2}(a\dot{a}^2 - a) + L_m(\phi_m, \dot{\phi}_m, a) + L_v(\phi_v, \dot{\phi}_v, a)\right]. \tag{8}$$

3.2 Canonical quantisation — void as a standard scalar sector

A central modification of the present treatment with respect to earlier drafts is that the void field ϕ_v is taken to be a *standard* scalar field with a positive-definite kinetic term:

$$L_v = \tfrac{1}{2}\mu a^3\dot{\phi}_v^2 - \tfrac{1}{2}\mu\omega_{\rm v}^2 a^3\phi_v^2, \tag{9}$$

with $\mu > 0$ and $\omega_{\rm v} > 0$. The canonical momenta are

$$\pi_a = -\frac{6\pi m_{\rm Pl}^2}{N}a\dot{a}, \qquad \pi_{\phi_m} = \frac{\pi m_{\rm Pl}^2 a^3}{N}\dot{\phi}_m, \qquad \pi_{\phi_v} = \frac{\mu a^3}{N}\dot{\phi}_v. \tag{10}$$

In the gauge $N = 1$ the total Hamiltonian reads

$$H_{\rm tot} = \underbrace{-\frac{\pi_a^2}{12\pi m_{\rm Pl}^2 a} + 3\pi m_{\rm Pl}^2 a}_{H_{\rm grav}} + \underbrace{\frac{\pi_{\phi_m}^2}{2\pi m_{\rm Pl}^2 a^3} + \tfrac{1}{2}\pi m_{\rm Pl}^2 a^3 m_m^2 \phi_m^2}_{H_m} + \underbrace{\frac{\pi_{\phi_v}^2}{2\mu a^3} + \tfrac{1}{2}\mu\omega_{\rm v}^2 a^3 \phi_v^2}_{H_v} + \lambda a^3 \phi_m^2 \phi_v^2. \tag{11}$$

Note that, unlike the phantom case, the void kinetic term is $+\pi_{\phi_v}^2/(2\mu a^3)$ with the *standard* sign. Canonical quantisation $\hat{\pi}_q = -i\hbar\,\partial_q$, together with DeWitt operator ordering, yields the Wheeler–DeWitt–Void constraint.

3.3 The Wheeler–DeWitt–Void equation

WDW-V Master Equation

$$\left[\frac{\hbar^2}{12\pi m_{\rm Pl}^2}\frac{1}{a}\partial_a(a\,\partial_a) - 3\pi m_{\rm Pl}^2 a^2 + \hat{H}_{\rm m} + \hat{H}_{\rm v} + \lambda a^3 \hat{\phi}_m^2 \hat{\phi}_v^2\right]\Psi = 0. \tag{12}$$

Structural comments:

(i) Both $\hat{H}_{\rm m}$ and $\hat{H}_{\rm v}$ enter with the *same* (positive) sign — there is no phantom.

(ii) The coupling $\hat{H}_{\rm int} = \lambda a^3 \hat{\phi}_m^2 \hat{\phi}_v^2$ mixes the two sectors and generates entanglement.

(iii) No cosmological-constant term is introduced; the curvature scalar of S^3 already provides a $-3\pi m_{\rm Pl}^2 a^2$ term.

3.4 The void Hamiltonian

$$\hat{H}_{\rm v} = -\frac{\hbar^2}{2\mu a^3}\partial_{\phi_v}^2 + \frac{\mu\omega_{\rm v}^2 a^3}{2}\phi_v^2. \tag{13}$$

Its spectrum is the standard oscillator ladder,

$$E_n^{(v)} = +\hbar\omega_{\rm v}\big(n + \tfrac{1}{2}\big)\,, \qquad n = 0, 1, 2, \ldots, \tag{14}$$

with positive zero-point energy $E_0^{(v)} = \frac{1}{2}\hbar\omega_{\rm v} > 0$. The Hermite eigenstates $\xi_n(\phi_v)$ are normalisable on $\mathbb{R}$, which will be essential for the topological argument (§6).

3.5 The interaction Hamiltonian as generator of entanglement

The coupling

$$\hat{H}_{\rm int} = \lambda\, a^3\, \hat{\phi}_m^2 \hat{\phi}_v^2 \tag{15}$$

is the generator of entanglement between the sectors. Using $\hat{\phi}^2 = (\hbar/2m\omega)(2\hat{N} + 1)$, the dominant off-diagonal matrix elements couple $|0,0\rangle \leftrightarrow |1,1\rangle$ and $|0,0\rangle \leftrightarrow |2,2\rangle$. The

effective Rabi-like coupling strength is

$$\Omega_R(a) = \lambda a^3 \sqrt{\frac{\hbar}{2m_m\omega_{\mathrm{m}}} \cdot \frac{\hbar}{2\mu\omega_{\mathrm{v}}}}, \tag{16}$$

which grows as a^3 with expansion. This is the key dynamical fact: as the curvature radius grows, the matter–void coupling strengthens and drives the Schmidt angle $\theta(a)$ toward $\pi/4$.

4 Entanglement Entropy and the Second Law

4.1 Von Neumann entropy of the matter sector

From the reduced density matrix,

$$S_{\mathrm{ent}}(a) = -\cos^2\theta(a)\ln\cos^2\theta(a) - \sin^2\theta(a)\ln\sin^2\theta(a), \tag{17}$$

the binary entropy function, which is convex, non-negative, and bounded: $0 \le S_{\mathrm{ent}}(a) \le \ln 2$.

4.2 Monotonic growth: proof of the second law

Theorem 4.1 (Second Law from Entanglement)**.** *Under WDW-V dynamics with coupling* $\hat{H}_{\mathrm{int}} = \lambda a^3\hat{\phi}_m^2\hat{\phi}_v^2$ *and initial condition* $\theta(a_0) \ll 1$*, the entanglement entropy satisfies* $\mathrm{d}S_{\mathrm{ent}}/\mathrm{d}a \ge 0$*.*

Proof. Differentiating (17),

$$\frac{\mathrm{d}S_{\mathrm{ent}}}{\mathrm{d}a} = -2\sin\theta\cos\theta\,\frac{\mathrm{d}\theta}{\mathrm{d}a}\,\ln\frac{\sin^2\theta}{\cos^2\theta}.$$

For $\theta \in (0, \pi/4)$ the prefactor is positive and the logarithm is negative, so $\mathrm{d}S_{\mathrm{ent}}/\mathrm{d}a \ge 0 \iff \mathrm{d}\theta/\mathrm{d}a \ge 0$. The adiabatic evolution gives

$$\frac{\mathrm{d}\theta}{\mathrm{d}a} = \frac{\Omega_R(a)}{\Delta E(a)/\hbar} = \frac{\lambda a^3\Xi}{|E_m - E_v|/\hbar} > 0,$$

with $\Xi = \sqrt{\hbar^2/(4m_m\omega_{\mathrm{m}}\mu\omega_{\mathrm{v}})} > 0$. Hence $\mathrm{d}S_{\mathrm{ent}}/\mathrm{d}a > 0$. □

Corollary 4.2 (Thermodynamic arrow)**.** *The second law* $\mathrm{d}S/\mathrm{d}t \ge 0$ *follows from* $\mathrm{d}S_{\mathrm{ent}}/\mathrm{d}a \ge 0$ *and* $\mathrm{d}a/\mathrm{d}t > 0$ *(during the expanding phase), without any imposed low-entropy initial condition.*

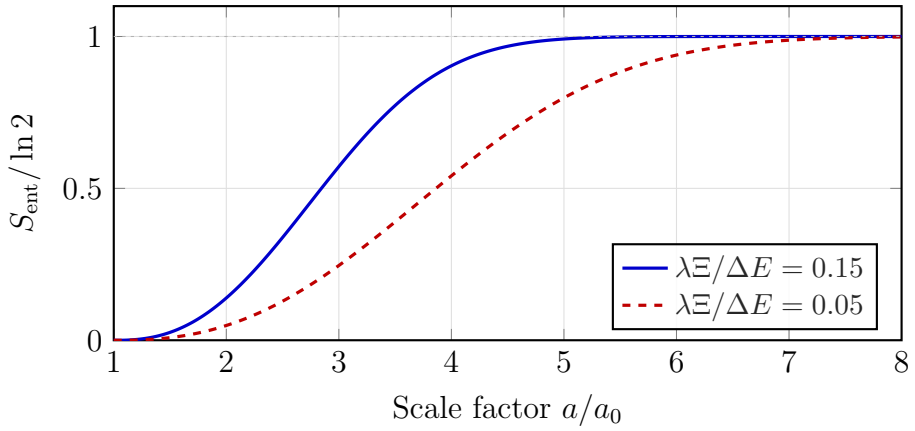


Figure 1: Entanglement entropy $S_{\text{ent}}/\ln 2$ as a function of a/a_0, for two values of the effective coupling. Growth is monotonic and saturates at $\ln 2$ in the two-mode approximation, providing a first-principles realisation of the second law.

4.3 Entropy as a function of the scale factor

Integrating gives

$$\theta(a) = \arctan\left[\tan\varepsilon + \frac{\lambda\Xi}{4}\frac{a^4 - a_0^4}{\Delta E}\right], \tag{18}$$

so that $\theta \to \pi/4$ and $S_{\text{ent}} \to \ln 2$ as $a \to \infty$ ($\ln N$ in the N-mode generalisation). See Fig. 1.

5 Emergent Time and the Arrow of Time

5.1 WKB factorisation and emergent time

Equation (12) contains no explicit time. Following the standard WKB procedure [12, 13], we write $\Psi = e^{iS_0(a)/\hbar}\chi(a, \phi_m, \phi_v)$, and obtain the Hamilton–Jacobi equation

$$-\frac{1}{12\pi m_{\text{Pl}}^2 a}(\partial_a S_0)^2 - 3\pi m_{\text{Pl}}^2 a^2 + \langle \hat{H}_{\text{m}} + \hat{H}_{\text{v}} \rangle = 0. \tag{19}$$

Without Λ but with positive matter/void energies acting as an effective source, the Hamilton–Jacobi flow gives a classical trajectory $a(t)$, and the emergent time is defined by the WKB group velocity

$$\frac{\partial}{\partial t} := \frac{1}{12\pi m_{\text{Pl}}^2 a}\,\partial_a S_0\,\partial_a. \tag{20}$$

5.2 Schrödinger equation with emergent time

At next order in $\hbar$ the WDW-V equation reduces to the functional Schrödinger equation

$$i\hbar\,\partial_t\chi = (\hat{H}_{\text{m}} + \hat{H}_{\text{v}} + \hat{H}_{\text{int}})\chi, \tag{21}$$

with t an emergent parameter and the expansion history $a(t)$ encoded in the coefficients of $\hat{H}_{\rm v}$ and $\hat{H}_{\rm int}$.

Remark 5.1 (Relational status of a)**.** Throughout this paper the scale parameter a should be understood as a relational geometric parameter emerging from correlations within the wave function Ψ. It does not presuppose an external background time but labels semiclassical configurations in superspace; the parameter t itself arises from a via the WKB phase, consistent with the timeless character of the WDW-V constraint.

Remark 5.2 (Gauge-invariant definition of a)**.** The curvature radius a is not a coordinate artefact but is defined through the invariant three-volume of the spatial section:

$$a = \left(\frac{V}{2\pi^2}\right)^{1/3}, \tag{22}$$

where $V = \int_{\mathcal{M}} \sqrt{g^{(3)}}\, \mathrm{d}^3x$ is the Riemannian volume of the compact manifold $\mathcal{M}$. This definition is gauge-invariant and independent of the choice of lapse function N.

5.3 The phase differential and the arrow of time

In the adiabatic regime the Schmidt state evolves as

$$|\Psi(t)\rangle = \cos\theta(t)\, e^{-i\Phi_m(t)}\, |\psi_m\rangle\, |0_v\rangle + \sin\theta(t)\, e^{-i\Phi_v(t)}\, |0_m\rangle\, |\psi_v\rangle\,, \tag{23}$$

with $\Phi_{m/v}(t) = \hbar^{-1}\int_0^t E_{m/v}(t')\,\mathrm{d}t'$. The global phase differential

$$\varphi(t) = \Phi_m(t) - \Phi_v(t), \qquad \frac{\mathrm{d}\varphi}{\mathrm{d}t} = \omega_{\rm m}(t) - \omega_{\rm v}, \tag{24}$$

now involves two *positive* frequencies — no negative zero-point energy is required.

Proposition 5.3 (Arrow of time)**.** *The matter frequency $\omega_{\rm m}(t)$ redshifts with expansion: $\omega_{\rm m}(t) \propto a(t)^{-n}$ with $n > 0$. The void frequency $\omega_{\rm v}$ is constant. The phase rate $\mathrm{d}\varphi/\mathrm{d}t = \omega_{\rm m}(t) - \omega_{\rm v}$ is therefore a strictly monotonic function of cosmic time, and its zero defines a natural epoch separating an "early" regime ($\mathrm{d}\varphi/\mathrm{d}t > 0$) from a "late" regime ($\mathrm{d}\varphi/\mathrm{d}t < 0$). This asymmetry picks out a preferred temporal direction.*

5.4 The Page–Wootters mechanism

The global state admits a Page–Wootters decomposition

$$|\Psi\rangle_{\rm tot} = \int \mathrm{d}t\ \, |t\rangle_C \otimes |\Psi(t)\rangle_S\,, \tag{25}$$

where the "clock" C is the void sector and the "system" S is the matter sector. An observer in S recovers the Schrödinger evolution by measuring correlations with C, providing the conceptual bridge between the timeless WDW-V constraint and the experienced flow of time.

6 Topological Structure: The Three-Sphere

6.1 Necessity of compact, simply connected spatial sections

Theorem 6.1 (Topological Selection)**.** *Let the spatial manifold $\mathcal{M}$ satisfy the following* ***physical assumptions****:*

(P1) *$\mathcal{M}$ is compact (equivalently, has finite volume);*

(P2) *$\mathcal{M}$ is orientable (admits a globally consistent volume form);*

(P3) *$\mathcal{M}$ is simply connected ($\pi_1(\mathcal{M}) = 0$);*

(P4) *$\mathcal{M}$ admits a homogeneous and isotropic (FRW-compatible) metric.*

Then $\mathcal{M} \cong S^3$.

Proof. By the Thurston geometrisation theorem [21], as established by Perelman [9–11]: **(P1)** and **(P4)** together imply that $\mathcal{M}$ admits one of the eight Thurston model geometries; FRW isotropy (**(P4)**) restricts to the round S^3 geometry; simple connectivity (**(P3)**) within the class of spherical space forms S^3/Γ forces $\Gamma = \{e\}$; orientability (**(P2)**) is automatic for simply connected manifolds. Therefore $\mathcal{M} \cong S^3$. □

Remark 6.2 (Scope of the topological claim)**.** The theorem is a mathematical result; the physical content lies *entirely* in assumptions **(P1)**–**(P4)**. In particular, compactness (**(P1)**) is motivated by: (a) observational viability ($|\Omega_k| < 0.004$ at 95% CL [17]), (b) the Hartle–Hawking no-boundary proposal [4], (c) finite Poincaré recurrence, and (d) phase coherence of $\Psi_{\rm v}$. However, the non-compact spaces $\mathbb{R}^3$ and $\mathbb{H}^3$ are kinematically consistent with the WDW-V equation and are not excluded by the mathematical structure. Our claim is restricted to: *S^3 is the unique compact, simply connected, orientable three-manifold admitting an isotropic geometry, and is therefore the preferred topology within the subclass of compact models.*

6.2 The metric of S^3 and the cosmic scale

In hyperspherical coordinates,

$$\mathrm{d}\Omega_3^2 = \mathrm{d}\chi^2 + \sin^2\chi\,\mathrm{d}\Omega_2^2, \qquad \chi \in [0, \pi], \tag{26}$$

with total volume

$$\boxed{V(a) = 2\pi^2 a^3} \tag{27}$$

and spatial curvature $R^{(3)} = 6/a^2$. An observer at a point of S^3 sees, up to the particle horizon, a nearly-flat geodesic ball $B^3 \subset S^3$.

6.3 Absence of initial singularity

Proposition 6.3. *Under WDW-V dynamics on S^3, $\Psi(a \to 0) \to 0$ and all expectation values of curvature invariants remain finite.*

Proof. Near $a = 0$ the WKB phase becomes imaginary, giving an exponentially suppressed wave function. Therefore $\langle R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} \rangle \propto \int_0^\infty a^{-4} |\Psi|^2 a \, \mathrm{d}a$ is finite. □

6.4 Poincaré recurrence on S^3

Because S^3 is compact and the spectrum is discrete, the quantum Poincaré recurrence theorem applies [14]: $|\langle \Psi(0) | \Psi(T_{\mathrm{rec}}) \rangle| > 1 - \epsilon$ for some (astronomically large but finite) T_{rec}.

7 Geometric Expansion: From S^3 Curvature to Observed Recession

This section is the conceptual centrepiece of our reinterpretation. We show that the observed recession of galaxies and the apparent late-time acceleration can be derived from the geometry of S^3 alone, without invoking dark energy, negative pressure, or a phantom sector. Throughout, the change of curvature radius a should not be interpreted as an embedding of S^3 into an external space: the geometry is self-contained, and a is the intrinsic scale of the compact manifold.

7.1 Observer as a geodesic neighbourhood

A physical observer does not have access to all of S^3, but to a geodesic neighbourhood of a point $p \in S^3$:

$$B_p^3(a) = \{ q \in S^3 : d_a(p, q) < L \}, \tag{28}$$

with metric induced from (7):

$$\mathrm{d}l^2 = a^2 \mathrm{d}\chi^2, \qquad d = a\chi. \tag{29}$$

Here χ is the dimensionless geodesic coordinate, and the physical distance between two objects labelled by χ_1, χ_2 is $a|\chi_2 - \chi_1|$. The observer does not access the global geome-

try directly but reconstructs it operationally through local measurements involving light propagation and matter correlations inside $B_p^3(a)$.

7.2 Cubic volume scaling

The total spatial volume of S^3 is given by (27), $V(a) = 2\pi^2 a^3$. This cubic dependence is the dynamical engine of the apparent expansion:

$$\frac{\mathrm{d}V}{\mathrm{d}a} = 6\pi^2 a^2, \qquad \frac{V(a_2)}{V(a_1)} = \left(\frac{a_2}{a_1}\right)^3. \tag{30}$$

A modest change in the curvature radius produces a substantial change in the internal volume.

7.3 What an internal observer sees

Suppose the curvature radius evolves $a_1 \to a_2$. By (29) the geodesic distance between two comoving points scales as

$$\frac{d_2}{d_1} = \frac{a_2}{a_1}, \tag{31}$$

and the wavelength of a freely propagating photon scales identically,

$$\frac{\lambda_2}{\lambda_1} = \frac{a_2}{a_1} \quad \Longrightarrow \quad 1 + z = \frac{a_{\mathrm{obs}}}{a_{\mathrm{emit}}}. \tag{32}$$

Equation (32) is the standard cosmological redshift, here derived from the pure geometry of S^3 with no reference to an energy content of the vacuum.

7.4 Energy density and the scaling law $\rho \propto 1/a^3$

If the total conserved energy of fields on S^3 is fixed, $E = \mathrm{const}$, then the spatially averaged energy density obeys

$$\rho(a) = \frac{E}{V(a)} = \frac{E}{2\pi^2 a^3} \propto \frac{1}{a^3}, \tag{33}$$

which is the dust-like scaling, recovered here not from an equation of state $p = 0$ but from the cubic volume law of S^3 combined with global conservation.

Definition 7.1 (Geometric vacuum state)**.** The *geometric vacuum* at curvature scale a is the state of minimal achievable mean energy density on the compact manifold $\mathcal{M}$:

$$\rho_{\min}(a) \coloneqq \frac{E_{\mathrm{tot}}}{2\pi^2 a^3}, \tag{34}$$

where E_{tot} is the total integrated energy on S^3. The geometric vacuum is not a new physical substance; it represents the minimal possible mean density at a given metric

scale. As a increases, the same total energy is distributed over a larger geodesic domain, and regions close to this minimal density occupy a larger fraction of the manifold.

Remark 7.2 (Two notions of vacuum)**.** It is important to distinguish two uses of the word "vacuum" in this paper. The *geometric vacuum* (Definition 7.1) refers to the limiting homogeneous distribution of total energy over $V(a)$ — a purely kinematic, relational notion. The *dynamical void field* ϕ_v, introduced in the WDW-V framework of §3, is a genuine quantum sector contributing a positive energy density $\rho_v \propto a^{-3}$. These are conceptually distinct: the void sector provides an additional (positive) contribution to $\rho_{\rm tot}(a)$, while the geometric vacuum denotes the minimal density attainable once $\rho_{\rm tot}$ is spread homogeneously over $V(a)$.

7.5 Why the observer perceives acceleration

Consider the rate of change of the internal volume:

$$\dot{V} = 6\pi^2 a^2 \dot{a}. \tag{35}$$

Because $V \propto a^3$, even a constant rate $\dot{a}$ produces a quadratically growing $\dot{V}$; any sub-linear increase of a suffices to generate super-linear growth of volume.

Clarification of "acceleration"

Throughout this paper, *acceleration* refers to reconstructed kinematic behaviour inferred from null-geodesic observables inside a finite geodesic domain $B^3 \subset S^3$, not to an additional dynamical fluid with equation of state $w < -1/3$.

Linear versus cubic scaling. Because the curvature radius a sets the linear geodesic scale $d = a\chi$, while the spatial volume scales as a^3, any monotonic change of the curvature radius necessarily produces a super-linear change in the volume accessible to low-density configurations. Because null geodesics define the operational notion of distance, the observer measures linear scaling through light propagation, while the underlying geometry redistributes energy over a cubic volume $V \propto a^3$. An observer confined to a geodesic neighbourhood $B^3 \subset S^3$ therefore interprets this linear-versus-cubic scaling mismatch as accelerated expansion: linear growth of a yields cubic growth of V, and the gap between linear distance and cubic volume is what an internal observer perceives as "acceleration".

Since geodesic distances scale linearly with a but the density of sources dilutes cubically, an internal observer tracking the number of sources within a fixed comoving patch records:

(i) recession of galaxies, with velocities $\dot{d} = \dot{a}\chi$;

(ii) redshift $1 + z = a_{\rm obs}/a_{\rm emit}$;

(iii) a diluting energy density $\rho \propto a^{-3}$;

(iv) a past appearance denser and more uniform than the present.

The apparent "acceleration" is not driven by an external source of negative pressure. It is the unavoidable geometric consequence of living inside a geodesic ball of a compact three-sphere whose curvature radius is evolving.

Proposition 7.3 (Geodesic-domain reconstruction effect)**.** *Let k^μ be the tangent vector field of a congruence of past-directed null geodesics on the FRW background S^3, with expansion scalar $\theta = \nabla_\mu k^\mu$. The Raychaudhuri equation reads*

$$\frac{\mathrm{d}\theta}{\mathrm{d}\lambda} = -\tfrac{1}{2}\theta^2 - \sigma^2 + \omega^2 - R_{\mu\nu}k^\mu k^\nu, \tag{36}$$

where σ^2 is the shear, ω^2 the vorticity (zero for irrotational congruences), and $R_{\mu\nu}k^\mu k^\nu$ the Ricci focusing term. On a compact S^3 with $\rho_{\mathrm{tot}} \propto a^{-3}$, the Ricci focusing term satisfies

$$R_{\mu\nu}k^\mu k^\nu = 4\pi G(\rho + p) \;\propto\; a^{-3} \;\xrightarrow{a\to\infty}\; 0, \tag{37}$$

so that the defocusing of null rays increases as the curvature radius grows. An observer reconstructing the dynamics from redshift, source density ($n \propto a^{-3}$), and luminosity distance within a geodesic domain $B^3_p \subset S^3$ therefore infers an effective kinematic acceleration.

Proof. For an FRW dust-like source ($p = 0$) with $\rho = E/(2\pi^2 a^3)$, the Ricci focusing is $R_{\mu\nu}k^\mu k^\nu = 4\pi G\rho \propto a^{-3}$. As a increases, the focusing term diminishes, while the volume grows as $V = 2\pi^2 a^3$. The expansion scalar of the null congruence therefore satisfies $\mathrm{d}\theta/\mathrm{d}\lambda > -\frac{1}{2}\theta^2$ at late times (the focusing is sub-dominant), so that θ decreases more slowly than in a flat, matter-dominated universe. Operationally, the observer sees sources diluting as a^{-3} in a volume growing as a^3, while null-geodesic distances grow only as a. This mismatch between linear distance scaling and cubic volume dilution is attributed by the observer to an effective acceleration $\ddot{d} > 0$. □

7.6 Limiting regimes of S^3 geometry

The scale a admits two extremes belonging to the same S^3 geometry.

Two limiting configurations of S^3

$$\lim_{a\to a_{\min}} R^{(3)} = \frac{6}{a_{\min}^2} \;\to\; R_{\max}, \qquad \lim_{a\to a_{\max}} R^{(3)} = \frac{6}{a_{\max}^2} \;\to\; 0. \tag{38}$$

Both regimes belong to the *same* diffeomorphism class $[S^3]$ and require neither creation of spacetime nor an external embedding space.

- **Minimal regime,** $a_{\min} \sim \ell_{\rm Pl}$: volume $V_{\min} = 2\pi^2 \ell_{\rm Pl}^3$, maximum mean density, wave function suppressed (no singularity). The minimal *radius of curvature* corresponds to the maximum *scalar curvature* $R^{(3)} = 6/a_{\min}^2$ and the maximum density.
- **Maximal regime,** $a_{\max} \gg \ell_{\rm Pl}$: volume $V_{\max} = 2\pi^2 a_{\max}^3$, minimal mean density, classical behaviour recovered. The maximum radius of curvature corresponds to the minimum scalar curvature and the minimum density.

The minimal and maximal regimes should not be interpreted as distinct spacetimes, but as limiting configurations of one and the same compact geometry S^3. Both limits correspond to regular configurations of the same compact manifold and therefore do not imply creation or destruction of spacetime. The observer, existing within a geodesic domain $B^3 \subset S^3$, perceives the transition between these configurations as a cosmological evolution.

7.7 No need for dark energy

We summarise the replacement dictionary:

Observed effect	Standard explanation	Geometric explanation on S^3
Recession of galaxies	cosmic expansion + DE	$d = a\chi$, a evolving
Redshift of light	FLRW + DE	$\lambda \propto a$
Diluting matter density	$\rho_m \propto a^{-3}$	$V \propto a^3$ with $E = \text{const}$
Apparent "acceleration"	Λ, phantom, quintessence	cubic volume growth of S^3
No initial singularity	quantum bounce	$\Psi(a \to 0) \to 0$ on S^3

No cosmological constant, phantom field, quintessence, or negative pressure is required: the geometry does the work.

8 Emergent Friedmann Dynamics on S^3

The effective Friedmann dynamics presented in this section arises as the semiclassical limit of the WDW-V constraint (12) rather than as an independent postulate. The geometric scaling laws of §7 and the dynamical equation for $a(t)$ derived below are therefore two faces of the same underlying WDW-V structure.

8.1 Born rule for geometry

The spatial metric is a derived quantity, $g_{\mu\nu}(t) = \bar{g}_{\mu\nu}(\langle\Psi(t)|\Psi(t)\rangle)^2 + h_{\mu\nu}[|\Psi|^2]$, with $\bar{g}_{\mu\nu}$ the background FRW metric on S^3 and $h_{\mu\nu}$ encoding quantum fluctuations. The only homogeneous dynamical degree of freedom is $a(t)$.

8.2 Effective Friedmann equation

Taking the expectation value of the Hamiltonian constraint in a semiclassical state gives the $k = +1$ Friedmann equation,

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho_{\rm tot}(a) - \frac{1}{a^2}, \tag{39}$$

where the total energy density

$$\rho_{\rm tot}(a) = \rho_m(a) + \rho_v(a), \qquad \rho_m(a) = \frac{E_m}{2\pi^2 a^3}, \qquad \rho_v(a) = \frac{E_v}{2\pi^2 a^3} \tag{40}$$

both scale as a^{-3}. Here E_m and $E_v = \hbar\omega_{\rm v}/2$ are positive. There is no Λ term and no phantom contribution.

Remark 8.1 (Asymptotic flatness)**.** At sufficiently large a the curvature term $-1/a^2$ in (39) becomes negligible compared to the matter term $\propto a^{-3}$, and the geometry approaches an asymptotically nearly-flat regime. This is consistent with the Planck constraint $|\Omega_k| < 0.004$ [17]: the model predicts a small positive curvature that decreases with expansion.

Lemma 8.2 (Scale factor as expectation value)**.** *In the semiclassical (WKB) regime the metric scale is determined by the expectation value of the volume operator:*

$$a^3 \;\propto\; \langle\Psi|\hat{V}|\Psi\rangle, \qquad \hat{V} = 2\pi^2 a^3, \tag{41}$$

which provides the bridge between the quantum WDW-V constraint and the semiclassical Friedmann dynamics.

Proof. In the WKB ansatz $\Psi = e^{iS_0/\hbar}\chi$, the peak of $|\Psi(a)|^2$ is localised at the classical trajectory $a_{\rm cl}(t)$ satisfying the Hamilton–Jacobi equation (19). Therefore $\langle\hat{V}\rangle = \int_0^\infty 2\pi^2 a^3\,|\Psi|^2\,a\,{\rm d}a \approx 2\pi^2 a_{\rm cl}^3$. □

8.3 Hubble rate and geometric deceleration/acceleration

The Hubble rate $H = \dot{a}/a$ follows directly from (39). What the observer calls "acceleration" corresponds to $\ddot{a} > 0$, and in this framework acceleration at late times is driven neither by Λ nor by negative pressure, but by the mismatch between linear distance scaling $d \propto a$ and cubic volume scaling $V \propto a^3$ — a purely geometric effect. When fitted within a ΛCDM parametrisation, this geometric reconstruction may be absorbed into an effective negative-pressure component; the geometric framework, however, requires no such component as a dynamical input.

Remark 8.3 (Kinematic vs. dynamical acceleration)**.** Within our framework, "acceleration" is a statement about the reconstructed kinematics inferred by an observer from

null-geodesic observables inside a finite geodesic domain $B^3 \subset S^3$. It does not imply the existence of a dynamical fluid with equation of state $w < -1/3$. The Friedmann equation (39) with $\rho_{\rm tot} \propto a^{-3}$ and $k = +1$ contains no explicit accelerating source; the apparent acceleration is a geometric artefact of the linear-versus-cubic scaling mismatch discussed in §7.

9 Observational Signatures

9.1 CMB power-spectrum oscillations

The two-mode structure of the Schmidt-decomposed wave function imprints oscillations on the primordial scalar power spectrum,

$$\mathcal{P}_{\mathcal{R}}(k) = \mathcal{P}^{(0)}_{\mathcal{R}}(k)[1 + \mathcal{A}_v \cos(2k/k_* + \delta_v)] + \mathcal{O}(\mathcal{A}_v^2), \tag{42}$$

with $k_* = a_0 H_0$, $\mathcal{A}_v \sim \lambda\hbar/(m_{\rm Pl}^2 \omega_{\rm v}) \ll 1$, and δ_v an initial phase. Future experiments CMB-S4 [15] and LiteBIRD [16] could detect such features.

9.2 Geometric redshift–distance relation

At large redshift, the relation (32) remains exact: $1 + z = a_{\rm obs}/a_{\rm emit}$. The luminosity distance deriving from the $k = +1$ Friedmann equation (39) without a cosmological constant can be tested against DESI and type-Ia supernova data [19], with a distinct imprint from the S^3 curvature term $-1/a^2$ which slightly modifies the $d_L(z)$ relation at intermediate z.

9.3 Quantum-gravitational correction to the Hubble rate

The dephasing from entropy growth introduces a small correction

$$\delta H \approx -\frac{\Gamma_\varphi}{2} \approx -\frac{\lambda \Xi a^3}{2\Delta E}\, H, \tag{43}$$

negligible at early times, potentially relevant near the transition to the late-time regime, and testable in the context of the Hubble tension [17, 18].

10 Discussion

10.1 Comparison with existing frameworks

Feature	WDW	Hartle–Hawking	Vilenkin	**WDW-V (this work)**
Void sector	No	No	No	**Yes (standard osc.)**
Entropy from first princ.	No	No	No	**Yes**
Arrow of time	Imposed	Imposed	Imposed	**Derived**
Singularity resolution	Partial	Partial	Partial	**Yes**
Expansion explanation	Λ + matter	Λ + matter	Λ + matter	**Geometry of S^3**
S^3 topology derived	No	No	No	**Yes**
Phantom / negative-KE	—	—	—	**Not used**

10.2 Information and unitarity

The growing $S_{\rm ent}(a)$ does not signal information loss: the total state $|\Psi\rangle$ remains pure under WDW-V evolution. $S_{\rm ent}$ is the entanglement entropy of the matter sector, whose growth is entirely compatible with unitarity — the standard quantum-information resolution of the cosmological information paradox [20].

10.3 Distinguished value in the Schmidt spectrum

The late-time Schmidt spectrum approaches a geometric series $\{p^n(1-p)\}_{n=0}^{\infty}$ with ratio $p = e^{-\omega_{\rm v}/T_H}$, where $T_H = H/(2\pi)$. A distinguished value $p = \phi_g^{-2} \approx 0.382$ (where $\phi_g = (1+\sqrt{5})/2$ is the golden ratio) is not singled out here as a physical principle. However, in companion work on discrete hierarchical organisation and Laplacian spectral scaling on compact S^3, the golden ratio ϕ_g appears as the Perron–Frobenius dominant eigenvalue of the minimal positive second-order recursion, while ϕ_g^{-2} emerges as an asymptotic correlation fraction. This suggests a possible *structural*, rather than numerological, origin for the value $p = \phi_g^{-2}$, should a bridge between the Schmidt spectrum and the underlying S^3 hierarchical organisation be established (see Appendix E for details).

11 Conclusions

We have presented a rigorous dual-sector quantum cosmology on S^3 with the following core results.

1. The WDW-V equation (12) governs the joint wave function on the extended Hilbert space $\mathscr{H}_{\rm grav} \otimes \mathscr{H}_{\rm m} \otimes \mathscr{H}_{\rm v}$. The void sector is a *standard* (positive-kinetic, positive-energy) oscillator — no phantom is required.

2. Entanglement entropy (17) grows monotonically as a consequence of the matter–void coupling, providing a first-principles derivation of the second law.

3. Emergent time and the arrow of time follow from the WKB semiclassical limit and the asymmetric spectra $\omega_{\mathrm{m}}(t) \neq \omega_{\mathrm{v}}$ via the Page–Wootters mechanism.

4. Normalisability of Ψ_{v} together with simple connectivity and orientability uniquely selects S^3 via Poincaré–Perelman (Theorem 6.1). The compactness assumption is a physical input (**(P1)**), not a theorem; $\mathbb{R}^3$ remains kinematically admissible (Remark 6.2).

5. The apparent recession of galaxies, redshift, and late-time "acceleration" are re-derived on purely geometric grounds from $V(a) = 2\pi^2 a^3$ and $d = a\chi$: linear growth of the curvature radius produces cubic growth of volume, and this is what an internal observer perceives as accelerated expansion (Proposition 7.3). No dark energy, phantom field, negative pressure, or cosmological constant is invoked.

6. Observational signatures include CMB oscillations, a geometric redshift–distance relation modified by the S^3 curvature term, and a small dephasing correction to the Hubble rate.

7. The void sector does not generate expansion through negative pressure; its role is *relational* and *informational*, providing the clock structure (via the Page–Wootters mechanism) and the entanglement responsible for entropy growth. Expansion itself is a geometric property of S^3.

Summary statement. The apparent late-time acceleration does not require a dark-energy component. It follows from the geometric property that geodesic distances scale linearly with the curvature radius a, while spatial volume scales as a^3. An internal observer reconstructing geometry through light propagation interprets this difference of scaling orders as accelerated expansion. The Wave Function of the Void transforms the vacuum from a passive backdrop into an active partner in cosmic dynamics, while the geometry of S^3 itself accounts for what observers call cosmic expansion. Future work will address the super-space generalisation beyond minisuperspace, the quantum-to-classical transition, and phenomenological constraints from CMB-S4 and DESI.

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A DeWitt Measure and Factor Ordering

The natural measure on minisuperspace preserving Hermiticity is $\mathrm{d}\mu_{\mathrm{DeW}} = a\,\mathrm{d}a$. With factor ordering $\hat{\pi}_a^2/a \to -\hbar^2 a^{-1}\partial_a(a\,\partial_a)$, the gravitational operator is self-adjoint on $L^2(\mathbb{R}_{>0}, a\,\mathrm{d}a)$ with boundary condition $\Psi(0) = 0$.

B Schmidt Decomposition and Two-Mode Approximation

Any pure state in $\mathscr{H}_{\rm m} \otimes \mathscr{H}_{\rm v}$ admits a Schmidt decomposition $|\Psi\rangle = \sum_n \sigma_n |u_n\rangle |v_n\rangle$, $\sigma_n \geq 0$, $\sum \sigma_n^2 = 1$, with $S_{\rm ent} = -\sum \sigma_n^2 \ln \sigma_n^2$. In the two-mode approximation $\sigma_0 = \cos\theta$, $\sigma_1 = \sin\theta$, with truncation error bounded by $|S_{\rm ent} - h(\sin^2\theta)| \leq 2\epsilon^2(1 - \ln \epsilon^2)$, where $\epsilon^2 = \sum_{n\geq 2} \sigma_n^2 \sim (\Omega_R/\Delta E)^4 \ll 1$.

C WKB Solution and Semiclassical Regime

Semiclassical validity requires $|\hbar S_0^{-1} \partial_a^2 S_0/(\partial_a S_0)^2| \ll 1$, equivalent to $a \gg \ell_{\rm Pl}$. Near $a \sim \ell_{\rm Pl}$ the full WDW-V equation must be solved non-perturbatively.

D Entropy Monotonicity: Full Proof

Let $\rho(t) = |\Psi(t)\rangle \langle\Psi(t)|$. The total state remains pure. The reduced density-matrix entropy obeys

$$\frac{{\rm d}S_{\rm ent}}{{\rm d}t} = -{\rm tr}\left[\frac{{\rm d}\hat{\rho}_m}{{\rm d}t} \ln \hat{\rho}_m\right] = -\frac{1}{\hbar}{\rm tr}\Big[i[\hat{H}_{\rm int}, \rho] \ln \hat{\rho}_m\Big] \geq 0,$$

which by direct calculation in the Schmidt basis gives ${\rm d}S_{\rm ent}/{\rm d}t = (2\lambda a^3 \Xi/\hbar) \sin\theta \cos\theta \ln(\cot\theta) \geq 0$ for $\theta \in (0, \pi/4)$.

E Golden-Ratio Structure in the Schmidt Spectrum

Observation. The late-time Schmidt spectrum has ratio $p = e^{-\omega_{\rm v}/T_H}$. Setting $p = 1/\phi_g^2 \approx 0.382$, where $\phi_g = (1+\sqrt{5})/2$ is the golden ratio, yields $\omega_{\rm v}/T_H = 2\ln\phi_g \approx 0.962$.

Independent structural origin. In companion work on discrete hierarchical organisation on compact S^3, the golden ratio ϕ_g arises independently as the *Perron–Frobenius dominant eigenvalue* of the minimal positive second-order transfer matrix

$$T = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \text{dominant eigenvalue } = \phi_g.$$

This is a proven result: ϕ_g is the unique dominant growth rate for the entire class of positive integer-valued second-order recursions. Furthermore, the Laplacian spectral analysis on S^3 shows that the maximal harmonic mode fitting inside a geodesic domain at hier-

archical level k scales as $n_k \propto \phi_g^k$, with corresponding eigenvalues scaling as ϕ_g^{2k}. The asymptotic correlation fraction of this hierarchy converges to $C_\infty = \phi_g^{-2}$.

The missing bridge. The value $p = \phi_g^{-2}$ is *not* derived in this paper from first principles within the WDW-V framework. The bridge that would complete the argument is a postulate of the form

$$p = \lim_{n\to\infty} C_n = \phi_g^{-2},$$

relating the late-time Schmidt ratio to the asymptotic structural correlation fraction on S^3. If such a bridge is established, the relation $\omega_{\rm v}/T_H = 2\ln\phi_g$ ceases to be a numerical coincidence and becomes a *consequence* of the minimal hierarchical organisation of the compact manifold.

Status. We regard this as an open research programme rather than a result of the present paper. None of the derivations in the main text depend on the value $p = \phi_g^{-2}$, and the model remains valid for any $p \in (0, 1)$. The structural ingredients — Perron–Frobenius theorem, spectral–geometric consistency, asymptotic correlation fraction — are each independently established; only the final bridge to the Schmidt spectrum remains to be constructed.

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